Angular Advanced Module - Monorepo’s

Peter Kassenaar
info@kassenaar.com

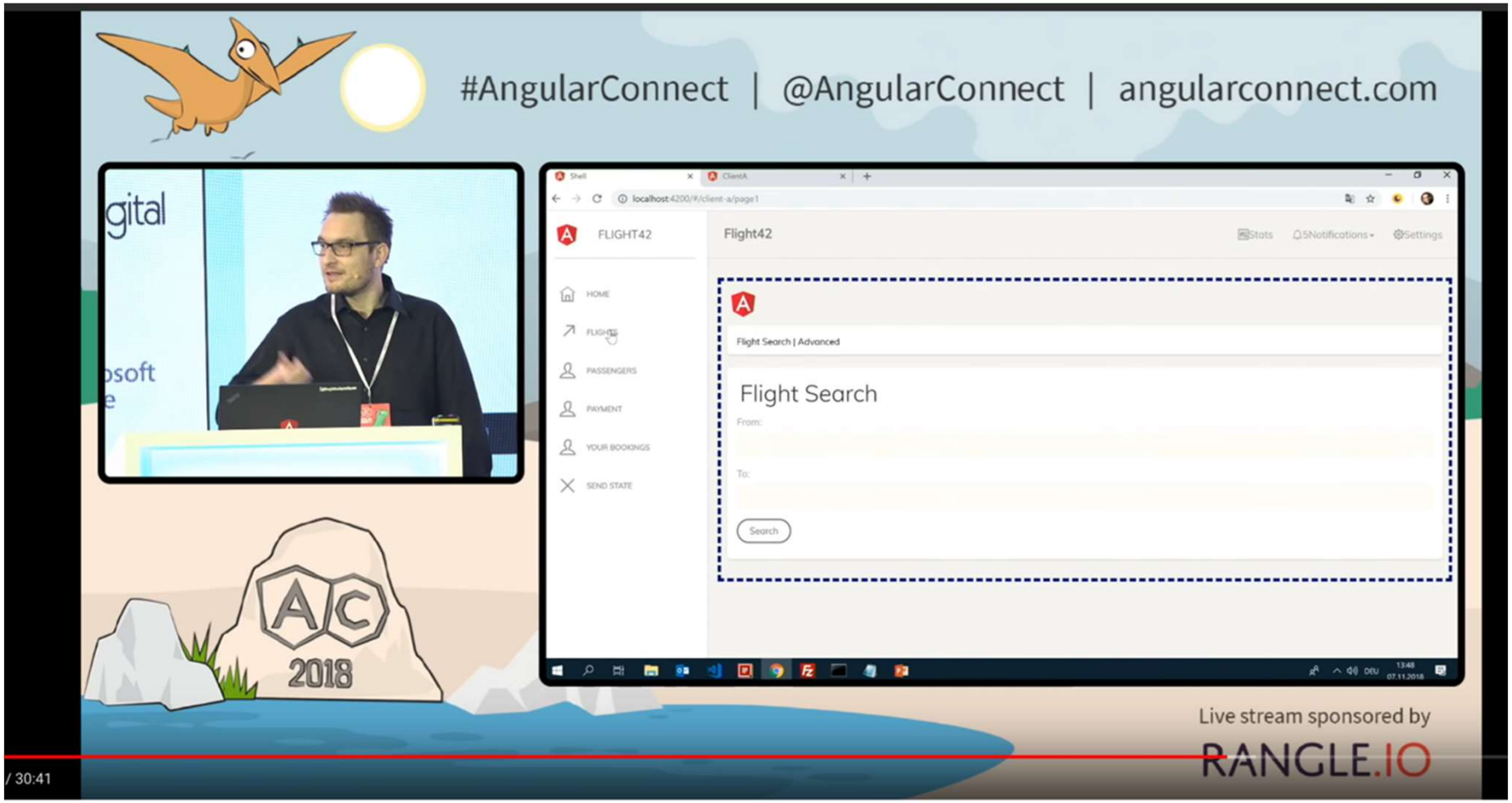
Angular in the Enterprise

- When? With (really) big(ger) applications
- Multiple solutions, examples:
  - Monorepo approach:

    https://github.com/PeterKassenaar/ng-monorepo
  - Micro-app approach:

    https://github.com/PeterKassenaar/ng-microfrontends

Manfred Steyer – Angular Connect



https://www.youtube.com/watch?v=YU-fMRs-ZYU

Code: https://github.com/PeterKassenaar/angular-microapp

Enterprise applications – multiple options

- There are always *multiple solutions*
- There is NOT one solution that is 'the best'

- Options:

1. NPM packages

   Publish your own packages/libraries to `npm`, so others can `npm install` them

2. Monorepo

   Multiple projects in one code base, optionally sharing code

3. Micro-apps / micro-frontends

   Multiple applications, not sharing code, optionally different techniques/frameworks

So basically...

NPM packages

Monorepo

Micro apps

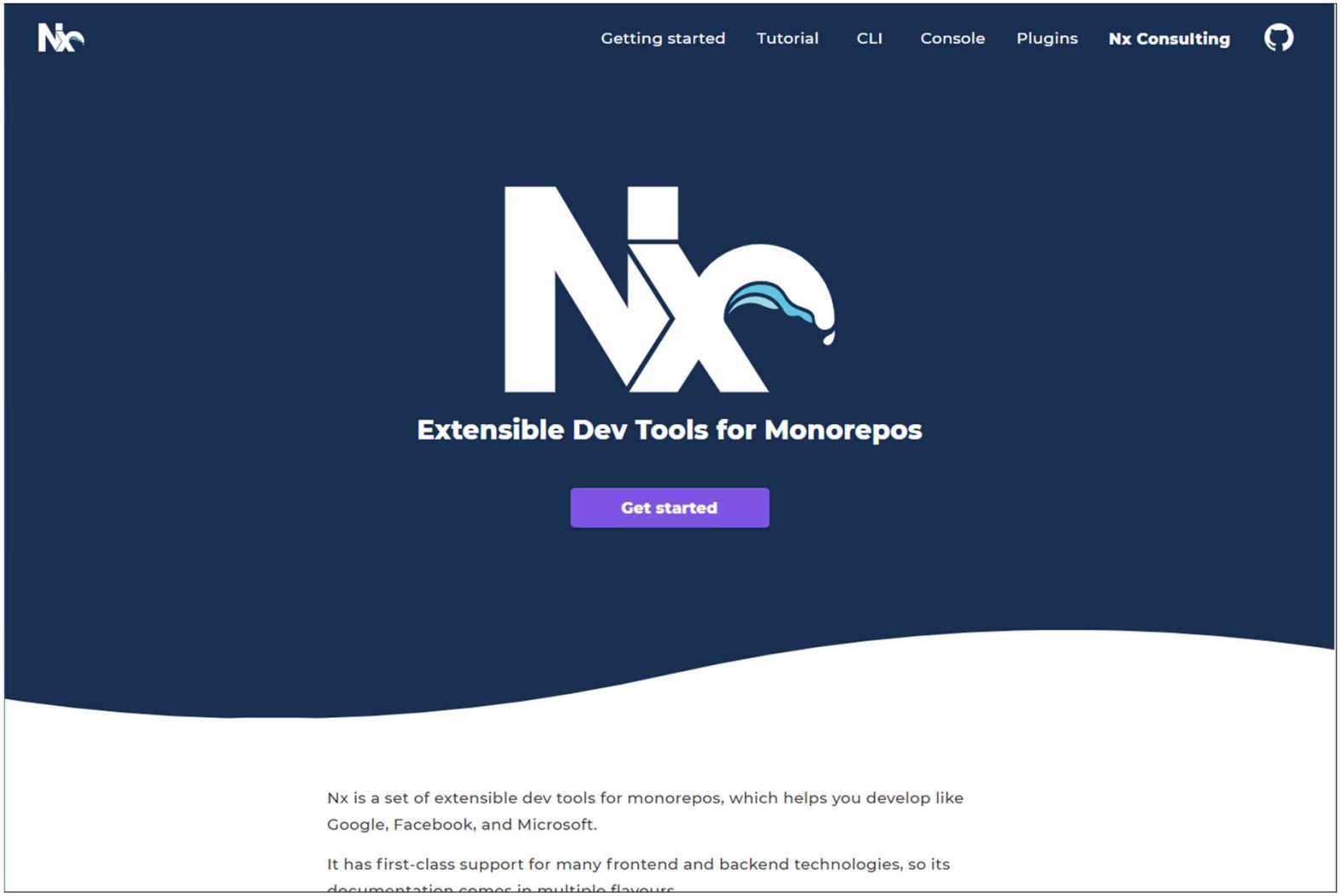
Monorepos

Multiple projects in one solution, optionally sharing code

What is a monorepo

- Often, bigger applications are not only split up into modules, but also in other applications
- Applications generally share a lot of code
  - Sharing components
  - Sharing services
  - Sharing pipes, other logic, etc…
- One (opiniated) solution – create a so called *monorepo*
- There are tools that also cover this
  - Nx Framework – free, open source, https://nx.dev/
  - Angular CLI as of V6.0.0+, https://cli.angular.io/

Nrwl extensions to create a monorepo



https://nx.dev/

Angular CLI – as of V6.0+

Which one to use?

- This is highly *opiniated*
- The work (roughly) the same
  - Creating a workspace
  - Creating applications and libraries inside that workspace
- Nx is also available for React
- Angular CLI is limited to (duh…) Angular
  - But…Nx is a wrapper around Angular CLI

mostly - Personal Preference!

We'll be using Angular CLI in this demo

ANGULAR FEATURES DOCS RESOURCES EVENTS BLOG Search

- Understanding Angular
- Developer Guides
- Best Practices
- Angular Tools
- Tutorials
- Release Information
- Reference
  - Conceptual Reference
  - CLI Command Reference
    - Overview
    - Usage Analytics
    - ng add
    - ng analytics
    - ng build
    - ng config
    - ng deploy
    - ng doc
    - ng e2e
    - ng extract-i18n
    - ng generate

CLI

ng generate

Generates and/or modifies files based on a schematic.

```
ng generate <schematic> [options]
ng g <schematic> [options]
```

Arguments

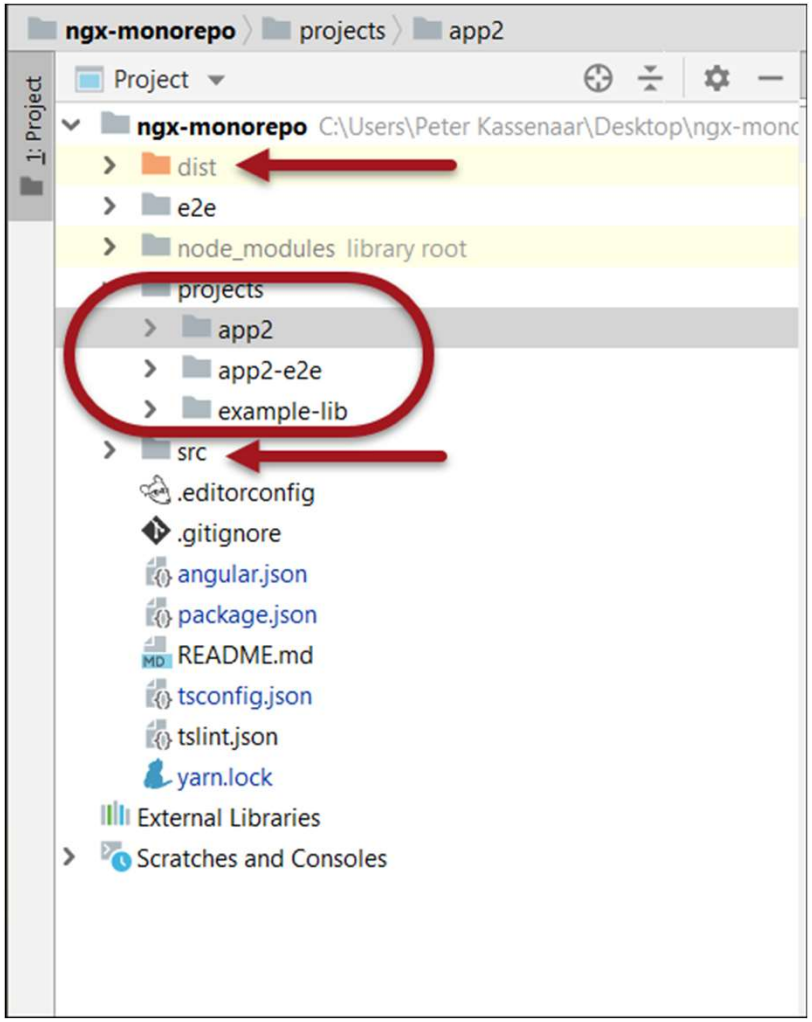
ARGUMENT	DESCRIPTION	VALUE TYPE
`<schematic>`	The schematic or collection:schematic to generate. This option can take one of the following sub-commands: • `app-shell` • `application` • `class` • `component` • `directive`	`string`

- ng generate
  - Arguments
  - Options
  - Schematic commands
    - app-shell
    - application
    - class
    - component
    - directive
    - enum
    - guard
    - interceptor
    - interface
    - library
    - module
    - pipe
    - resolver
    - service
    - service-worker

Inside an Angular monorepo

- One `/node_modules`
- One main `package.json`
- `angular.json`, describing all the projects in the workspace
- Optional - One root app, in `/src` folder
- One `/projects` folder, containing:
  - A folder for every project
  - Libraries
  - Applications
  - This folder is created upon the first creation of a `library` or `application`
- `/dist` folder, holding the compiled Angular Packages that needs to be shared

Structure/architecture



Credits: Angular in Depth

https://blog.angularindepth.com/creating-a-library-in-angular-6-87799552e7e5

High level overview of the steps

1. Create the container monorepo using `ng new <mono-repo-name>`
   - We call this folder a *workspace*
2. Create *without* the main app: `ng new <name> --create-application=false`!
   - https://blog.angularindepth.com/angular-workspace-no-application-for-you-4b451afcc2ba

```
ng new ng-monorepo --create-application=false

C:\Users\Peter Kassenaar\Desktop>ng new ng-monorepo --create-application=false
? Would you like to add Angular routing? No
? Which stylesheet format would you like to use? CSS
CREATE ng-monorepo/angular.json (135 bytes)
CREATE ng-monorepo/package.json (1262 bytes)
CREATE ng-monorepo/README.md (1027 bytes)
CREATE ng-monorepo/tsconfig.json (435 bytes)
CREATE ng-monorepo/tslint.json (2824 bytes)
CREATE ng-monorepo/.editorconfig (246 bytes)
CREATE ng-monorepo/.gitignore (576 bytes)
warning karma > log4js > circular-json@0.5.9: CircularJSON is in maintenance onl
y, flatted is its successor.
[##################################----------------------------] 10717/19987
```

Create (shared) lib

- Generate the (shared) library
  - `cd ng-monorepo`
  - `ng generate library shared-lib --prefix=sl` (or some other prefix)
  - Angular also updates the global `angular.json`, `package.json` and `tsconfig.json`
- Always use custom prefixes on libraries and projects
  - distinguish components and services!

```
ng g library shared-lib --prefix=sl

C:\Users\Peter Kassenaar\Desktop>cd ng-monorepo

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>ng g library shared-lib --prefix=sl

CREATE projects/shared-lib/karma.conf.js (968 bytes)
CREATE projects/shared-lib/ng-package.json (159 bytes)
CREATE projects/shared-lib/package.json (140 bytes)
CREATE projects/shared-lib/tsconfig.lib.json (726 bytes)
CREATE projects/shared-lib/tsconfig.spec.json (246 bytes)
CREATE projects/shared-lib/tslint.json (245 bytes)
CREATE projects/shared-lib/src/test.ts (700 bytes)
CREATE projects/shared-lib/src/public_api.ts (171 bytes)
CREATE projects/shared-lib/src/lib/shared-lib.module.ts (240 bytes)
CREATE projects/shared-lib/src/lib/shared-lib.component.spec.ts (650 bytes)
CREATE projects/shared-lib/src/lib/shared-lib.component.ts (266 bytes)
CREATE projects/shared-lib/src/lib/shared-lib.service.spec.ts (349 bytes)
CREATE projects/shared-lib/src/lib/shared-lib.service.ts (138 bytes)
UPDATE angular.json (1276 bytes)
UPDATE package.json (1439 bytes)
UPDATE tsconfig.json (573 bytes)
[##########################################################-----] 11004/11980
```

angular.json

```
1   {
2     "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
3     "version": 1,
4     "newProjectRoot": "projects",
5     "projects": {
6       "shared-lib": {"root": "projects/shared-lib"...}
41    },
42    "defaultProject": "shared-lib"
43  }
```

```
17      "lib": [
18        "es2018",
19        "dom"
20      ],
21      "paths": {
22        "shared-lib": [
23          "dist/shared-lib"
24        ],
25        "shared-lib/*": [
26          "dist/shared-lib/*"
27        ]
28      }
29    }
30  }
```

`tsconfig.json` – added paths, so we can import shared stuff easily later on

Create first app in the monorepo

- Generate the (first) application
  - `ng generate application <application-name>`
  - Again, `angular.json` and `package.json` are updated with the new project

```
Opdrachtprompt

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>ng g application dashboard
CREATE projects/dashboard/src/favicon.ico (5430 bytes)
CREATE projects/dashboard/src/index.html (296 bytes)
CREATE projects/dashboard/src/main.ts (372 bytes)
CREATE projects/dashboard/src/polyfills.ts (3571 bytes)
CREATE projects/dashboard/src/test.ts (642 bytes)
CREATE projects/dashboard/src/styles.css (80 bytes)
CREATE projects/dashboard/src/assets/.gitkeep (0 bytes)
CREATE projects/dashboard/src/environments/environment.prod.ts (51 bytes)
CREATE projects/dashboard/src/environments/environment.ts (662 bytes)
CREATE projects/dashboard/browserslist (388 bytes)
```

Create a shared component

- Of course you can create multiple components
  - `ng generate component <component-name> --project shared-lib`
  - Use the `--project` flag to tell the CLI what project you want to add the component to
- Give the component some UI

```
Opdrachtprompt

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>ng g c test  --project shared-lib
CREATE projects/shared-lib/src/lib/test/test.component.html (23 bytes)
CREATE projects/shared-lib/src/lib/test/test.component.spec.ts (614 bytes)
CREATE projects/shared-lib/src/lib/test/test.component.ts (260 bytes)
CREATE projects/shared-lib/src/lib/test/test.component.css (0 bytes)
UPDATE projects/shared-lib/src/lib/shared-lib.module.ts (310 bytes)

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>
```

Export the component

- Add it to the exports array of the `shared-lib.module.ts`

  `exports: […, TestComponent]`

- Update the `public_api.ts` file to make the class available
  - Don't forget!

```
/*
 * Public API Surface of shared-lib
 */

export * from './lib/shared-lib.service';
export * from './lib/shared-lib.component';
export * from './lib/shared-lib.module';
export * from './lib/test/test.component';
```

Build the library

- In order to use the component(s) from a shared library, it must be build
  - `ng build shared-lib`
  - A `\dist` folder is created
  - This folder is already mentioned in `tsconfig.json`. Verify this!

```
Opdrachtprompt

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>ng build shared-lib
Building Angular Package
Building entry point 'shared-lib'
Compiling TypeScript sources through ngc
Bundling to FESM2015
Bundling to FESM5
Bundling to UMD
Minifying UMD bundle
Copying declaration files
Writing package metadata
Removing scripts section in package.json as it's considered a potential security
 vulnerability.
Built shared-lib
Built Angular Package!
 - from: C:\Users\Peter Kassenaar\Desktop\ng-monorepo\projects\shared-lib
 - to:   C:\Users\Peter Kassenaar\Desktop\ng-monorepo\dist\shared-lib

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>
```

Using the library in the application

- Import the library module in the application `app.module.ts`
  - in our example: `dashboard.module.ts`
  - Remove the path your IDE might add to the AutoImport

```
import { AppComponent } from './app.component';
import {SharedLibModule} from '../../../shared-lib/src/lib/shared-lib.module';

@NgModule({
  declarations: [
    AppComponent
  ],
  imports: [
    BrowserModule,
    SharedLibModule
  ],
  providers: [],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

```
import { AppComponent } from './app.component';
import {SharedLibModule} from 'shared-lib';

@NgModule({
  declarations: [
    AppComponent
  ],
  imports: [
    BrowserModule,
    SharedLibModule
  ],
  providers: [],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

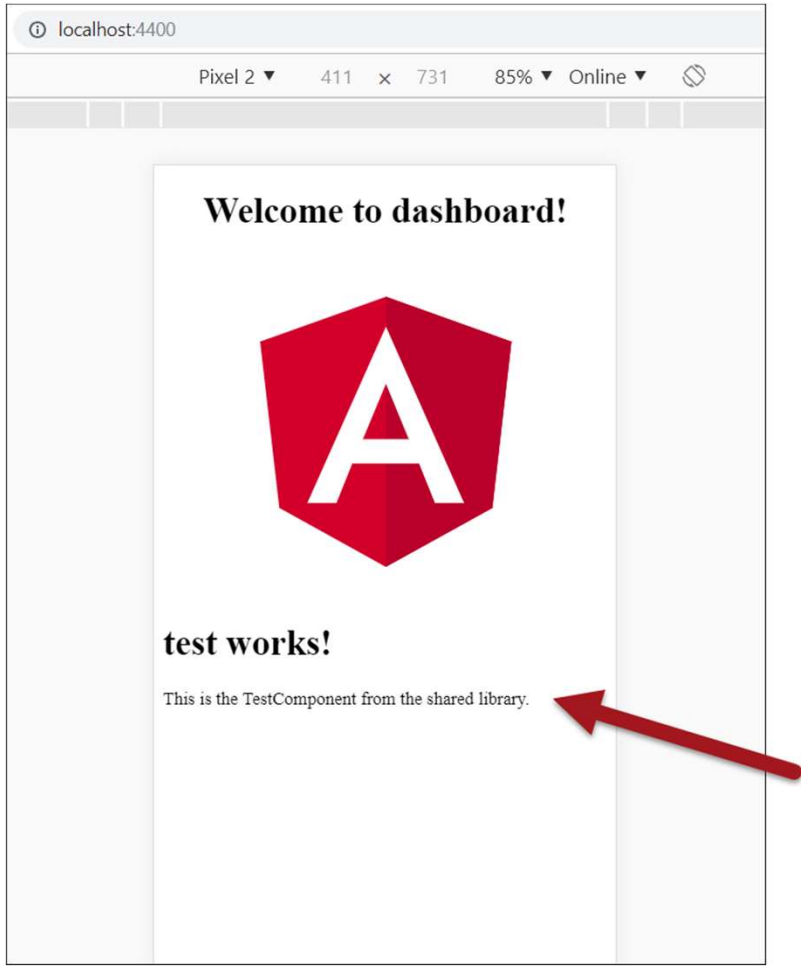
Using the shared component

- Use the selector of the component from the shared library as normal

```
<div style="text-align:center">
   …
</div>
<sl-test></sl-test>
```

Running the application

- Use the `ng serve` command
  - Use the `--project=<application-name>` to open the correct project
  - You can use additional flags as needed



- If you update your library contents, you need to rebuild it!
  - Write a script for that. For example

```
"scripts": {
   …
  "build_lib": "ng build shared-lib"
},
```

Exporting a service

- Create a service the regular way
  - `ng generate service <service-name> --project=<project-name>`
- You need to make sure to export your service from the module
- But is only needs to be loaded once!
  - Use a `.forRoot()` method on the module

```
Opdrachtprompt

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>ng g s shared/services/user --proje
ct shared-lib
CREATE projects/shared-lib/src/lib/shared/services/user.service.spec.ts (323 byt
es)
CREATE projects/shared-lib/src/lib/shared/services/user.service.ts (133 bytes)

C:\Users\Peter Kassenaar\Desktop\ng-monorepo>
```

Creating a `.forRoot()`

```
@NgModule({
  declarations: [SharedLibComponent, TestComponent],
  imports: [
  ],
  exports: [SharedLibComponent, TestComponent]
})
export class SharedLibModule {
  static forRoot(): ModuleWithProviders {
    return {
      ngModule: SharedLibModule,
      providers: [ UserService ]
    };
  }
}
```

- Remember to export the service from `public_api.ts`
- Remember to rebuild the library

Using the shared service

- In your application, update the module to use `SharedLibModule.forRoot()`
- Inject the Service in the component where you want to use it
- Use the servicemethods as normal

```
@NgModule({
  …
  imports: [
    SharedLibModule.forRoot()
  ],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

Component

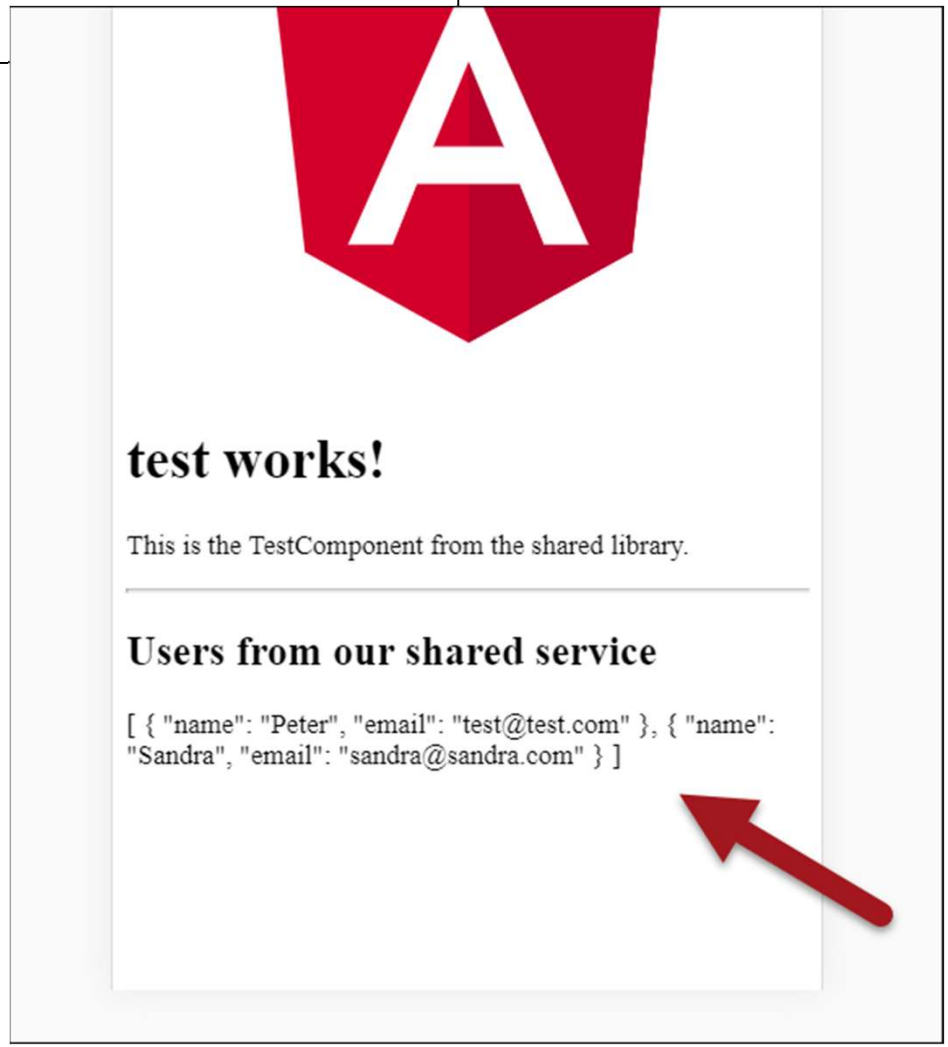
```
import {Component, OnInit} from '@angular/core';
import {User, UserService} from 'shared-lib';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.css']
})
export class AppComponent implements OnInit {
  title = 'dashboard';
  users: User[];

  constructor(private userService: UserService) {
  }

  ngOnInit(): void {
    this.users = this.userService.getUsers();
  }
}
```

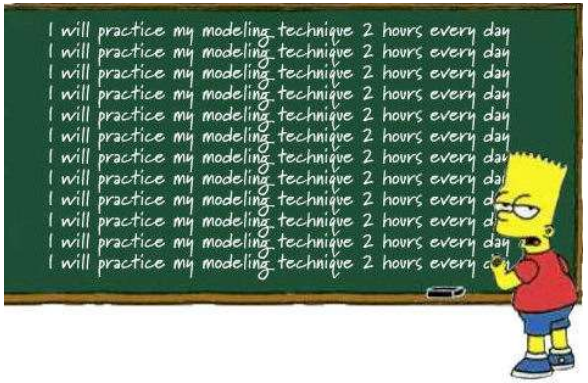
```
<hr>
<h2>Users from our shared service</h2>
{{ users | json }}
```



Workshop

- Create a new application inside the Monorepo
- Import the shared library
- Use/show the `<sl-test>` component from the shared lib inside your new project
- Build and run the new project
- Optional: create a new shared component in the lib.
  - Export / use the new component inside your project
  - Use the exported shared service from the library

Example: github.com/PeterKassenaar/ng-monorepo



Verdict on monorepo

- PRO:
  - Same version of Angular
  - Clear responsibilities per project/application
  - Share code and share services
- CON
  - Harder to use shared state / store
  - All the pro’s, are actually also cons! (depending on your project)
  - Checkout: you always get the complete monorepo- unless working with *Github subtrees*

More info

Info on npm packages, monorepo’s and micro apps

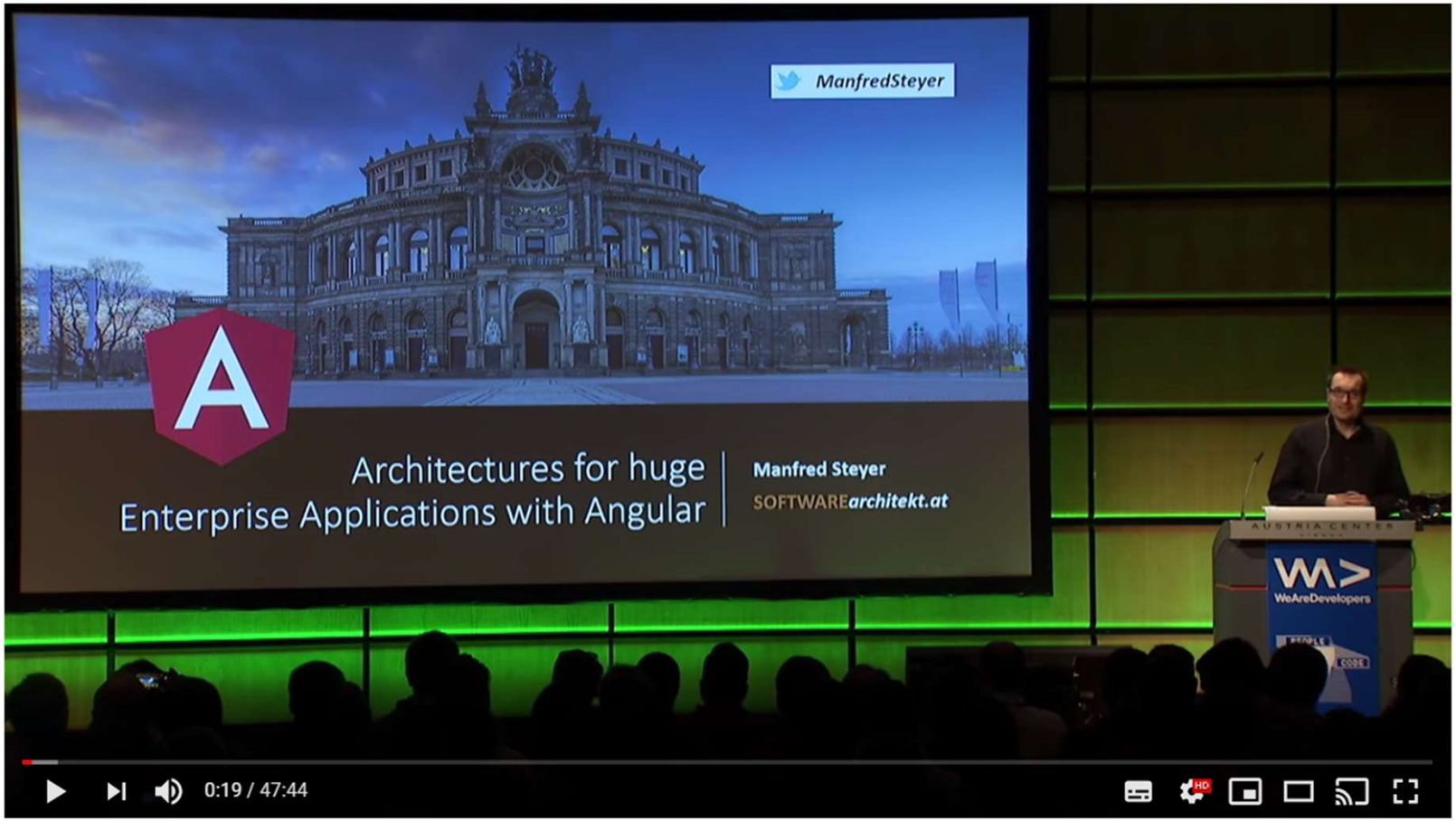
Info on...

NPM packages

Monorepo

Micro apps

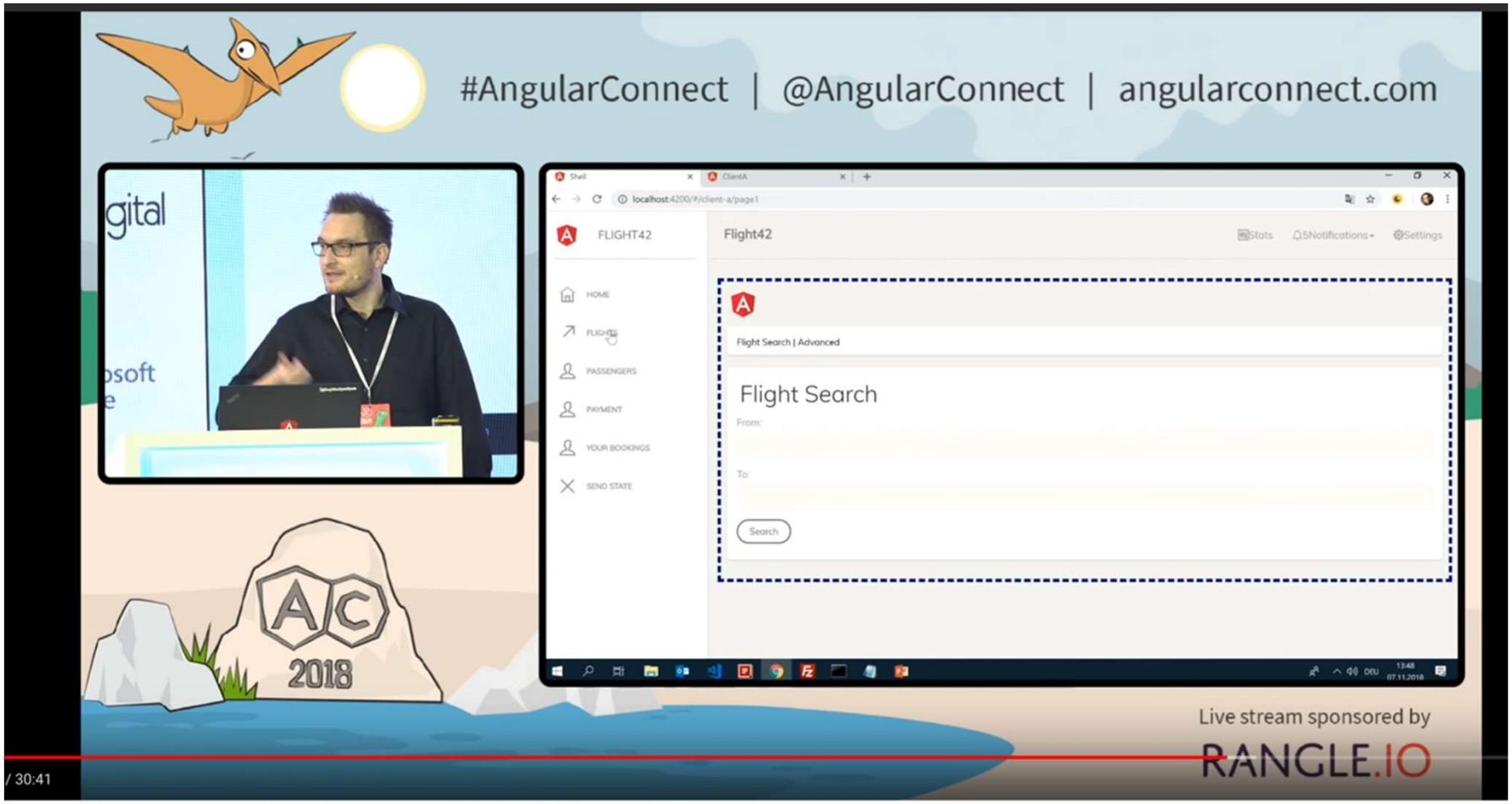
Talks on Angular Monorepo's



#WeAreDevs
Architectures for Huge Angular Based Enterprise

- https://www.youtube.com/watch?v=q4XmAy6_ucw

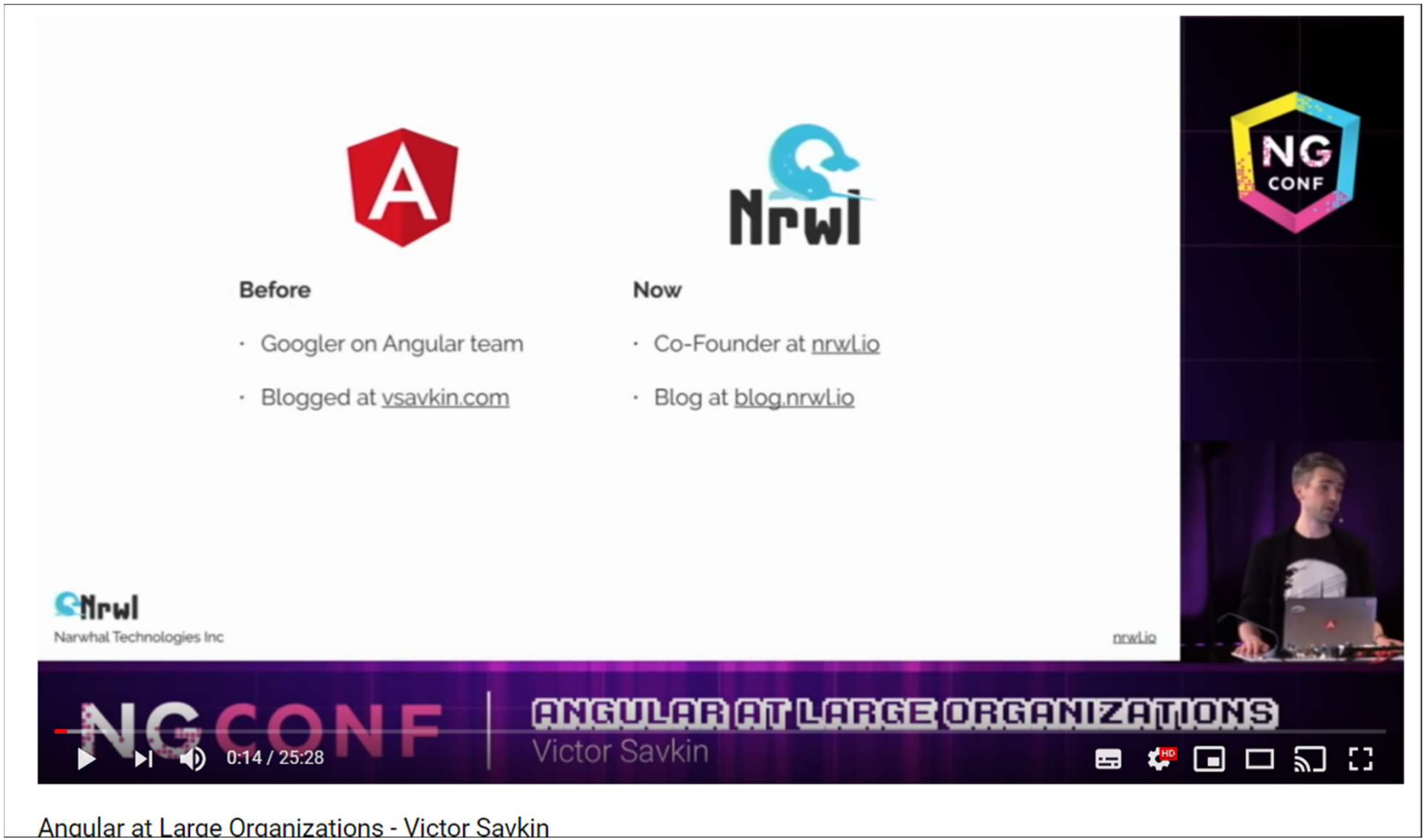
Manfred Steyer – Angular Connect



https://www.youtube.com/watch?v=YU-fMRs-ZYU

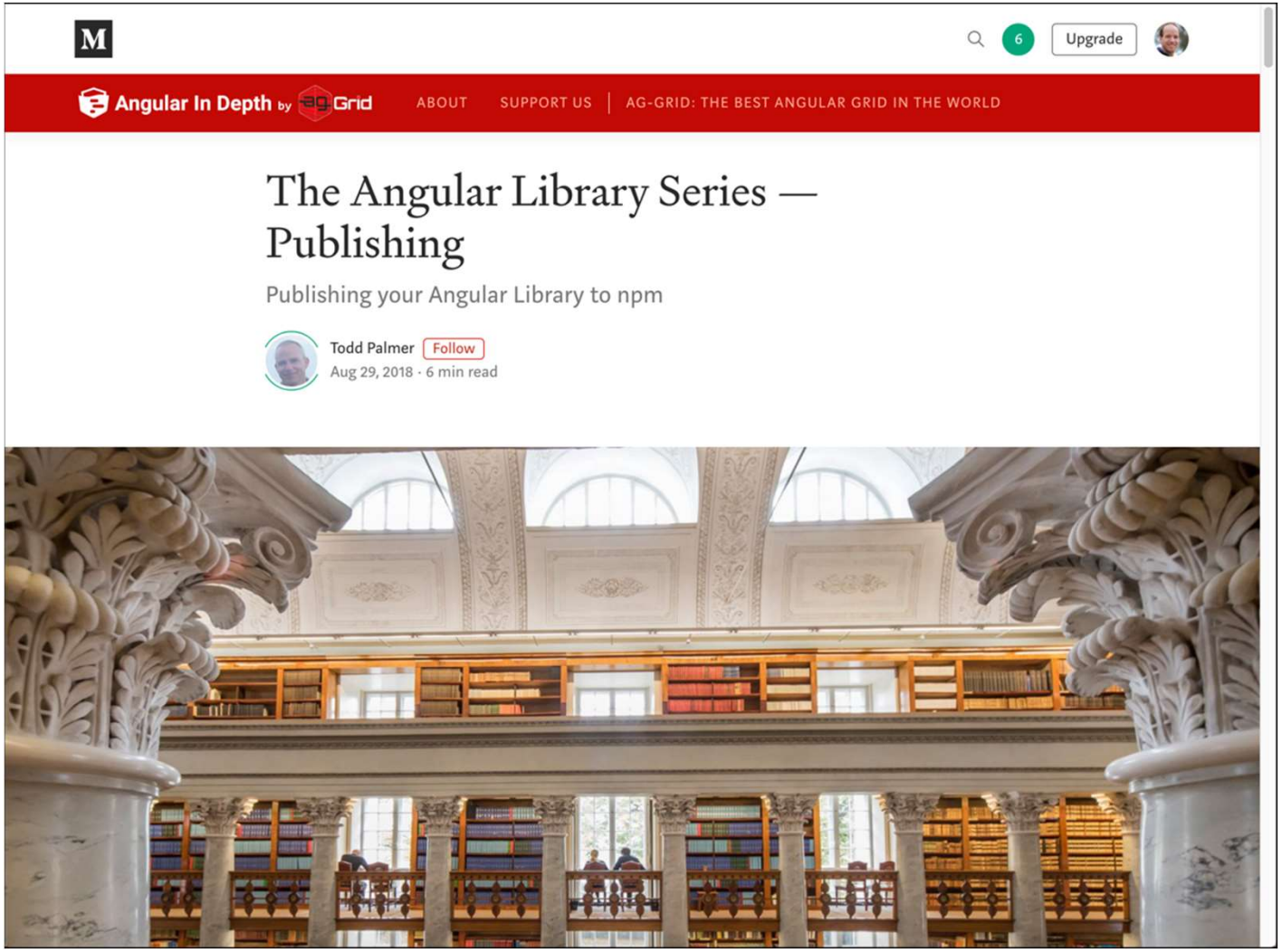
Code: https://github.com/PeterKassenaar/angular-microapp

Victor Savkin – creator of Nx



- https://www.youtube.com/watch?v=piQ0EZhtus0

Publishing your library to npm



https://blog.angularindepth.com/the-angular-library-series-publishing-ce24bb673275

Implementing micro apps in Angular

https://medium.com/bb-tutorials-and-thoughts/how-to-implement-micro-frontend-architecture-with-angular-e6828a0a049c